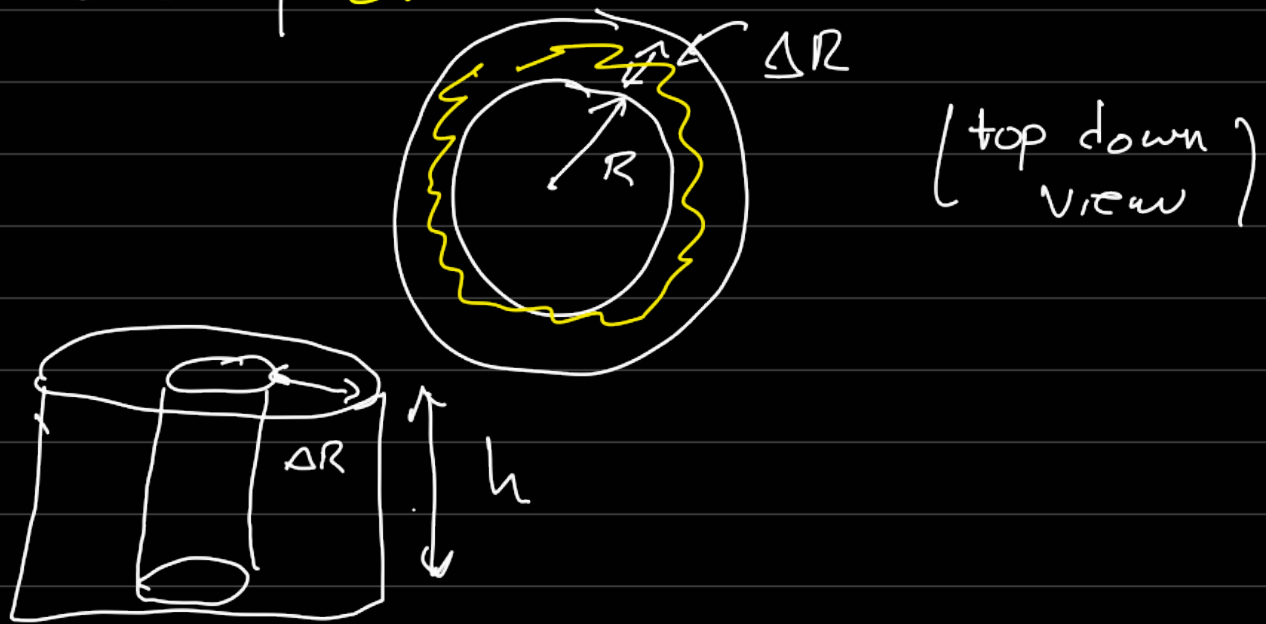


- $M(<R) = \frac{RV^2}{G}$

means that $V(R)^2 = \frac{GM(<R)}{R}$

- Let's build a simple model

$$dm = \rho dV$$



$$dV = (2\pi R dR) h$$

$$\int dm = \int_0^R \rho(\tilde{R}) 2\pi \tilde{R} h d\tilde{R}$$

* Recall we had IMF

$$V(R)_t^2 = \frac{G h \int_0^R 2\pi \tilde{R} \rho(\tilde{R}) d\tilde{R}}{R}$$

- Find $\rho(R)$ (not easy):

$V_t^2 \sim \frac{1}{R}$ (outside of Buldge)
 → However Post this initial radius

$$V_t^2(R_b) \sim \frac{1}{R_b}$$

the V_t does not continue to drop

- Disk dominates at ground

$\sim 3 \text{ kpc}$
 V_t : minimum

Sun: $\sim 8 \text{ kpc}$

$R_{MW} \approx 13 \text{ kpc}$

• Past 3 kpc V_t rises again

• $\sim 5 \text{ kpc}$ on constant

$\Rightarrow$ outer parts of galaxy have not settled into a state of lowest energy

$\sim 20 \text{ kpc}$: Dominated by dark matter

Stellar density profile:

$$n(R, z, s) = n(0, 0, s) e^{-R/h_R(s)} e^{-|z|/h_z(s)}$$

Polytropes & Isothermals:

(Kent 1986)

(Begeman et al 1991)

• Determine the mass of MW:

$\sim 30-50 \text{ kpc}$ $(V(R) \approx \text{const.})$

• Seen by HII regions emissions and
21cm of HI

* Mass-to-Luminosity Ratio:

$\rightarrow$ Difference between galaxy mass predicted

by its luminosity (Power output in light)
and the mass predicted by rotational
velocities: $\left(v^2 = \frac{GM}{R} \right)$

• Surface-mass-Density:
 $\Sigma(R)$

• Both models same for $\sim 30 \text{ kpc}$

Divide: $\text{SMD} : \frac{\Sigma(R)}{L(R)}$

• Increases Monotonically as we near the
edges:

$$\frac{M}{L} = 2 : M = 2L$$

$$L_{\text{MW}, B} \approx 2.3 \times 10^{10} L_{\odot, B}$$

• If MW has all solar like stars:

$$M_{\text{MW}} \approx 2.3 \times 10^{10} M_{\odot}$$

$$L_{\text{MW}} / M_{\text{MW}} \approx L_{\odot, B} / M_{\odot}$$

$$\ast \text{ IMF: } \sum f(M) dM: \quad \frac{M}{L_B} \sim 10^{-3} \frac{M_{\odot}}{L_{\odot}}$$

$$L \propto M^{-3.5}$$

$$\bullet \text{ Around the Sun: } \left\langle \frac{M}{L_B} \right\rangle \sim 4 \frac{M_{\odot}}{L_{\odot}}$$

$$\approx 170,000 \text{ kg m}^{-2}$$

$$V(R) = \sqrt{\frac{GM(R)}{R}}$$

* Mass in hydrostatic eg: $\rho \approx \frac{1}{R^2}$

$$dm = 4\pi\rho^2 dR \approx 4\pi dR \sim 4\pi R$$

$$V(R) \approx \text{const.}$$

